

August 28, 2026

RE: Staff Software Engineer, Backend, Cresta

Dear Hiring Manager,

I am applying for the Staff Software Engineer, Backend role at Cresta. The opportunity to shape distributed backend systems for enterprise AI agents closely matches my work leading platform architecture, high-volume data infrastructure, and production engineering in a regulated environment. I bring hands-on system design and delivery experience alongside the staff-level responsibility of guiding engineers and aligning technical decisions across teams.

At RBC, I architected an institution-wide observability and data platform built with Vector, Fluent Bit, Kafka, and Logstash. It ingests 150 TB per day across more than 10,000 systems and replaces point-to-point vendor integrations with configurable data paths. I also defined OpenTelemetry schemas, collection pipelines, and SDK onboarding guidance adopted by more than 20 engineering teams. This work required careful reasoning about throughput, failure isolation, schema contracts, operability, and long-term platform evolution, the same fundamentals behind reliable, high-volume backend services.

I also designed and led the 0-to-1 production delivery of a patent-pending Node.js and TypeScript architecture platform. I wrote 99.6% of its production code and all core logic for service matching, real-time rendering, and rules-based auto-certification. Building the platform in air-gapped, security-constrained environments sharpened my approach to API design, secure integrations, deployment reliability, and pragmatic architectural trade-offs. I am now extending it through ASB Assist, an LLM + RAG interface under development that converts natural-language requirements into compliant architecture diagrams and enterprise service recommendations. That work gives me direct experience collaborating at the boundary between AI capabilities and the deterministic systems that make them usable and governable.

Earlier, as HealthCard's first full-stack engineer, I built Node.js/Express APIs and deployed containerized workloads on AWS EKS with HIPAA-compliant data handling and audit logging. The team grew to more than 20 engineers before acquisition. Across these roles, I have owned systems from design through production, built CI/CD and observability into delivery, and worked across cloud, security, product, and architecture concerns.

Today I lead five engineers through technical planning, ADRs, code and design reviews, mentorship, sprint execution, and production handoff. I would welcome the chance to bring that combination of hands-on backend engineering, distributed-systems judgment, and cross-team leadership to Cresta as it scales the runtime infrastructure behind its AI platform.

Thank you for your consideration. I look forward to discussing how I can contribute to Cresta's backend architecture and production excellence.

Best regards,
Dhruv Doshi

SKILLS

- **Backend & APIs:** TypeScript, JavaScript, Python, Go, SQL, Node.js, REST APIs, GraphQL, service-oriented architecture
- **Distributed Data Systems:** Kafka, Airflow, Elasticsearch, Redis, ELK Stack, event streaming, schema design
- **Cloud & Infrastructure:** AWS (EKS, Lambda, API Gateway, RDS, S3, Kinesis), GCP, Kubernetes, Docker, Terraform, Helm, CI/CD
- **Reliability & Observability:** OpenTelemetry, distributed tracing, SLOs, Prometheus, alerting pipelines, production ownership
- **AI Systems:** LLMs, RAG, LLM orchestration, embeddings, prompt engineering, OpenAI API, Bedrock, AI governance
- **Technical Leadership:** distributed systems architecture, API design, ADRs, code reviews, mentorship, cross-team roadmaps

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect Nov 2022 – Present
 - **Large-Scale Distributed Data Platform:** Architected an institution-wide platform using Vector, Fluent Bit, Kafka, and Logstash that ingests 150 TB/day across 10,000+ systems; designed vendor-agnostic data paths that replaced point-to-point integrations.
 - **0-to-1 Backend Platform Ownership:** Designed and led production delivery of a patent-pending Node.js/TypeScript platform; wrote 99.6% of production code and all core logic for real-time rendering, service matching, and rules-based auto-certification.
 - **Reliability & Developer Enablement:** Defined OpenTelemetry schemas, collection pipelines, and SDK onboarding guides adopted by 20+ engineering teams, improving production observability while enabling vendor changes through configuration.
 - **AI Runtime Collaboration:** Building ASB Assist, an LLM + RAG interface that translates natural-language requirements into compliant architecture diagrams and service recommendations with security patterns and auditability.
 - **Architecture & Mentorship:** Lead 5 engineers through design, code reviews, ADRs, sprint delivery, and production handoff; align product, risk, security, and architecture stakeholders on cross-team technical decisions.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Cloud-Native Backend Delivery:** Built the product from scratch across Node.js/Express REST APIs, React/TypeScript, Hyperledger Fabric, and AWS EKS with HIPAA-compliant data handling and audit logging; team scaled to 20+ engineers before acquisition.
 - **Containerization & Release Engineering:** Containerized workloads on Kubernetes and built a Hardhat/Foundry CI/CD pipeline, reducing deployment cycles from days to under 30 minutes.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
IT Developer / Software Developer Co-op Apr 2022 – Nov 2022
 - **Production Platform Modernization:** Migrated a federal tax platform from Apache Struts to React/TypeScript, contributed to APIs serving millions of Canadians, and established automated SonarQube quality gates.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Systems Research & Teaching:** Developed call-stack decision algorithms for compiler/runtime research and supported Algorithms, Software Development, and graduate Advanced Cloud Computing courses.

PROJECTS

- **DCS-BBN – Decentralized Cloud Storage:** Designed a peer-to-peer storage network spanning thousands of nodes with Solidity, Hyperledger Fabric, and zk-SNARK verification; published at the Springer ICMR International Conference on Mobile Computing.

EDUCATION

- **Dalhousie University** Halifax, Canada
Master of Applied Computer Science, GPA: 3.9/4.0 Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020